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import pandas as pd

import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.linear_model import LogisticRegression

from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay


# Load the dataset

df = pd.read_csv("diabetes.csv")


# Remove missing values

df.dropna(inplace=True)


# Separate features and target

x = df.drop("Outcome", axis=1)

y = df["Outcome"]


# Split the dataset into training and testing sets

x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)


# Logistic Regression without feature scaling

model_2 = LogisticRegression(max_iter=10000)

model_2.fit(x_train, y_train)


pred_2 = model_2.predict(x_test)


accuracy_2 = accuracy_score(y_test, pred_2)

precision_2 = precision_score(y_test, pred_2)

recall_2 = recall_score(y_test, pred_2)

f1_score_2 = f1_score(y_test, pred_2)


print("\n---- Logistic Regression Without Feature Scaling ----")

print("\n---- MODEL PERFORMANCE ----")

print("Accuracy : ", accuracy_2)

print("Precision : ", precision_2)

print("Recall : ", recall_2)

print("F1 Score : ", f1_score_2)
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# Confusion matrix

cm = confusion_matrix(y_test, pred_2)

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=["No Diabetes", "Diabetes"]
)

disp.plot()

plt.title("Without Feature Scaling")

plt.show()


# Apply feature scaling

scaler = StandardScaler()

x_train = scaler.fit_transform(x_train)

x_test = scaler.transform(x_test)


# Logistic Regression with feature scaling

model_1 = LogisticRegression(max_iter=10000)

model_1.fit(x_train, y_train)


pred_1 = model_1.predict(x_test)


accuracy_1 = accuracy_score(y_test, pred_1)

precision_1 = precision_score(y_test, pred_1)

recall_1 = recall_score(y_test, pred_1)

f1_score_1 = f1_score(y_test, pred_1)


print("\n---- Logistic Regression With Feature Scaling ----")

print("\n---- MODEL PERFORMANCE ----")

print("Accuracy : ", accuracy_1)

print("Precision : ", precision_1)

print("Recall : ", recall_1)

print("F1 Score : ", f1_score_1)


# Confusion matrix

cm = confusion_matrix(y_test, pred_1)

disp = ConfusionMatrixDisplay(
    confusion_matrix=cm,
    display_labels=["No Diabetes", "Diabetes"]
)

```

```
disp.plot()

plt.title("With Feature Scaling")

plt.show()

---- Logistic Regression Without Feature Scaling ----

---- MODEL PERFORMANCE ----
Accuracy : 0.7467532467532467
Precision : 0.6379310344827587
Recall : 0.6727272727272727
F1 Score : 0.6548672566371682

---- Logistic Regression With Feature Scaling ----

---- MODEL PERFORMANCE ----
Accuracy : 0.7532467532467533
Precision : 0.6491228070175439
Recall : 0.6727272727272727
F1 Score : 0.6607142857142857
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